

Insurance Toolkits API Documentation

Overview

This API provides access to ITK's quoting and underwriting tools.

Authenticate using the X-API-KEY header.

Base URL

Use the API host provided by your ITK contact, for example:

- Production: `https://api.insurancetoolkits.com`

All examples below use `{{BASE_URL}}`.

Authentication

Send your API key in the X-API-KEY request header.

X-API-KEY: <your_api_key>
Content-Type: application/json

Authentication behavior

- Invalid Authentication -> 401

Usage tracking

- Usage is tracked monthly for requests to `/quoter/*`
- `/questionnaire/*` requests are currently not counted in quote usage counters

Error Format

Errors are normalized to:

```
{
  "status_code": 401,
  "error": "API key is missing."
}
```

Common statuses:

- 200 success
- 400 validation/business-rule errors
- 401 authentication failures

Quoter Endpoints

POST /quoter/

Runs quote + underwriting and returns sorted quote results.

Request body (common fields)

- sex ("Male" or "Female") - required
- state (2-letter state code) - required
- age (number) or month + day + year - required
- tobacco (string, e.g. "None") - recommended
- toolkit (FEX, TERM, MEDSUPP, HI, IUL) - optional (default FEX)
- faceAmount or premiumAmount - optional
- coverageType - optional; accepted values include ULTIMATE_PREFERRED, SUPER_PREFERRED, LEVEL, STANDARD, GRADED, MODIFIED, EASY_ISSUE, GUARANTEED; aliases Graded/Modified, Limited Pay, and SPWL are supported in quote flows
- paymentType - optional; one of Bank Draft/EFT, Direct Express, Credit Card, Debit Card (default Bank Draft/EFT)
- underwritingItems - optional array

Optional fields by toolkit/use case:

- effectiveDate (must not be in the past)
- zipcode, countyName, planCode (MedSupp flows)
- term, returnOfPremium (Term flows)
- riders, hhdToggle, days, dailyBenefit, carriersFilter

Minimal example

```
curl -X POST "{{BASE_URL}}/quoter/" \
-H "X-API-KEY: {{API_KEY}}" \
-H "Content-Type: application/json" \
-d '{
  "sex": "Male",
  "age": 60,
  "state": "TX",
  "tobacco": "None",
  "coverageType": "ULTIMATE_PREFERRED",
  "paymentType": "Bank Draft/EFT",
  "underwritingItems": []
}
```

```

"tobacco": "None",
"toolkit": "FEX",
"coverageType": "Level",
"faceAmount": 10000,
"underwritingItems": []
},

```

Success response (full fields)

Top-level object:

- quotes (List[QuoteResult])
- excluded (List[ExcludedCompany])

QuoteResult fields:

- company (String)
- monthly (String) - monthly premium, formatted
- yearly (String) - yearly premium, formatted
- med_supp_policy_fee (List[Number]) - MedSupp only, otherwise usually [0]
- med_supp_hhds (List[Number]) - MedSupp household discount percentages
- med_supp_hhds_applied_monthly (List[Object]) - each item has:
 - discount_percentage (Integer)
 - discounted_rate (String)
- med_supp_hhds_applied_yearly (List[Object]) - each item has:
 - discount_percentage (Integer)
 - discounted_rate (String)
- med_supp_plan_code (String) - MedSupp plan code (e.g. F, G, N)
- med_supp_pricing_type (String) - MedSupp pricing type
- am_best_rating (String) - MedSupp AM Best rating when available
- face_amount (String) - formatted face amount used in the result
- by_premium (Boolean) - whether quote was run from premium input
- tier (Integer) - internal coverage tier
- tier_name (String) - carrier-specific coverage label for the tier
- why (List[String]) - underwriting/logic reasons
- warning (String | null) - warnings, if any
- addnd (String | null) - accidental death monthly text
- addnd_yearly (String | null) - accidental death yearly text
- full_comp (Boolean) - commission full-comp flag
- comp_percent_lower (String) - compensation reduction percent
- compensation_text (String) - compensation explanation text
- eapp_link (String) - carrier eApp URL or #
- logo (String) - logo URL
- plan_name (String) - display plan name
- plan_info (List[String]) - display plan info lines
- limited_pay (List[LimitedPayResult])
- is_ssb_company (Boolean)
- effective_date (String | null) - MedSupp effective date when present
- is_spwl_quote (Boolean)

LimitedPayResult fields:

- constraint (String) - e.g. 10-pay, 20-pay
- monthly (String)
- yearly (String)
- addnd_monthly (String | null)
- addnd_yearly (String | null)
- hhds_applied_monthly (List[Object]) - discount_percentage, discounted_rate
- hhds_applied_yearly (List[Object]) - discount_percentage, discounted_rate
- med_supp_hhds (List[Integer])
- face_amount (String)
- by_premium (Boolean)
- warning (String | null)
- effective_date (String | null)

ExcludedCompany fields:

- name (String)
- why (List[String])

```

{
  "quotes": [
    {
      "company": "Sentinel",
      "monthly": "42.42",
      "yearly": "493.30",
      "med_supp_policy_fee": [0],
      "med_supp_hhds": [0],
      "med_supp_hhds_applied_monthly": [],
      "med_supp_hhds_applied_yearly": [],
      "med_supp_plan_code": "",
      "med_supp_pricing_type": "",
      "am_best_rating": "",
      "face_amount": "10,000.00",
      "by_premium": false,
      "tier": 0,
      "tier_name": "New Vantage I (Level)",
      "why": [],
      "warning": null,
      "addnd": null,
      "addnd_yearly": null,
      "full_comp": true,

```

```

    "comp_percent_lower": "0",
    "compensation_text": "",
    "eapp_link": "https://example.com/eapp",
    "logo": "https://insurancetoolkits.com/img/landing/companies/sentinel.jpg",
    "plan_name": "New Vantage I (Level)",
    "plan_info": [],
    "limited_pay": [
      {
        "constraint": "20-pay",
        "monthly": "50.31",
        "yearly": "585.00",
        "addnd_monthly": "50.31",
        "addnd_yearly": "585.00",
        "hhds_applied_monthly": [],
        "hhds_applied_yearly": [],
        "med_supp_hhds": [],
        "face_amount": "10,000.00",
        "by_premium": false,
        "warning": null,
        "effective_date": null
      }
    ],
    "is_ssb_company": false,
    "effective_date": null,
    "is_spwl_quote": false
  }
],
"excluded": [
  {
    "name": "Some Company",
    "why": ["Reason excluded"]
  }
]
}

```

GET /quoter/compare/

Returns companies and available coverage types for compare mode.

Query params

- state - required (2-letter state code)
- toolkit - required (FEX, TERM, MEDSUPP, HI, IUL)

Example

```

curl "{{BASE_URL}}/quoter/compare/?state=TX&toolkit=FEX" \
-H "X-API-KEY: {{API_KEY}}"

```

Success response (full fields)

Top-level object:

- data (List[CompareQuoteResult])

CompareQuoteResult fields:

- all fields from QuoteResult (see POST /quoter/ above)
- plus is_by_premium (Boolean) on each item

```

{
  "Sentinel": ["New Vantage I (Level)", "New Vantage II (Graded)"]
}

```

POST /quoter/compare/

Runs side-by-side compare quoting for selected company and values.

Request body

- company - required
- state - required
- toolkit - required
- coverageType - required in FEX compare flows; must match one of the coverage labels returned by GET /quoter/compare/ for the selected company/state
- demographic/quote fields similar to /quoter/
- values - required array
 - type: FACE_AMOUNT or PREMIUM (some flows may use PLAN)
 - value: numeric/string amount

Example

```

curl -X POST "{{BASE_URL}}/quoter/compare/" \
-H "X-API-KEY: {{API_KEY}}" \
-H "Content-Type: application/json" \
-d '{
  "sex": "Male",
  "age": 60,
  "state": "TX",
  "tobacco": "None",
  "toolkit": "FEX",
  "company": "Sentinel",
  "coverageType": "New Vantage I (Level)",
  "values": [

```

```

    {"type": "FACE_AMOUNT", "value": 10000},
    {"type": "PREMIUM", "value": 65}
  ],
},

```

Response

```

{
  "data": [
    {
      "company": "Sentinel",
      "monthly": "69.93",
      "yearly": "813.10",
      "med_supp_policy_fee": [
        0
      ],
      "med_supp_hhds": [
        0
      ],
      "med_supp_hhds_applied_monthly": [],
      "med_supp_hhds_applied_yearly": [],
      "med_supp_plan_code": "",
      "med_supp_pricing_type": "",
      "am_best_rating": "",
      "face_amount": "10,000.00",
      "by_premium": false,
      "tier": 0,
      "tier_name": "New Vantage I (Level)",
      "why": [],
      "warning": "",
      "addnd": null,
      "addnd_yearly": null,
      "full_comp": true,
      "comp_percent_lower": "0",
      "compensation_text": "",
      "eapp_link": "https://sslco.admin-portal.org/",
      "logo": "https://insurancetoolkits.com/img/landing/companies/sentinel.jpg",
      "plan_name": "New Vantage I (Level)",
      "plan_info": [],
      "limited_pay": [
        {
          "is_spwl_quote": false,
          "monthly": "124.38",
          "yearly": "1,446.30",
          "addnd_monthly": null,
          "addnd_yearly": null,
          "hhds_applied_monthly": [],
          "hhds_applied_yearly": [],
          "med_supp_hhds": [],
          "face_amount": "10,000.00",
          "by_premium": false,
          "warning": null,
          "effective_date": null,
          "constraint": "10-pay"
        }
      ],
      "is_ssb_company": false,
      "effective_date": null,
      "is_spwl_quote": false,
      "is_by_premium": false
    },
    {
      "company": "Sentinel",
      "monthly": "63.24",
      "yearly": "735.29",
      "med_supp_policy_fee": [
        0
      ],
      "med_supp_hhds": [
        0
      ],
      "med_supp_hhds_applied_monthly": [],
      "med_supp_hhds_applied_yearly": [],
      "med_supp_plan_code": "",
      "med_supp_pricing_type": "",
      "am_best_rating": "",
      "face_amount": "9,000.00",
      "by_premium": true,
      "tier": 0,
      "tier_name": "New Vantage I (Level)",
      "why": [],
      "warning": "Sentinel only allows face amounts in increments of $1,000. Rate calculated at face amount of $9,000. The resulting p",
      "addnd": null,
      "addnd_yearly": null,
      "full_comp": true,
      "comp_percent_lower": "0",
      "compensation_text": "",
      "eapp_link": "https://sslco.admin-portal.org/",
      "logo": "https://insurancetoolkits.com/img/landing/companies/sentinel.jpg",
      "plan_name": "New Vantage I (Level)",
      "plan_info": [],
      "limited_pay": [
        {
          "is_spwl_quote": false,
          "monthly": "63.70",
          "yearly": "740.65",
          "addnd_monthly": null,
          "addnd_yearly": null,
          "hhds_applied_monthly": [],
          "hhds_applied_yearly": [],
          "med_supp_hhds": [],
          "face_amount": "5,000.00",
          "by_premium": true,

```

```

        "warning": "Sentinel only allows face amounts in increments of $1,000. Rate calculated at face amount of $5,000. The res",
        "effective_date": null,
        "constraint": "10-pay"
    }
},
"is_ssb_company": false,
"effective_date": null,
"is_spwl_quote": false,
"is_by_premium": true
}
]
}
}

```

GET /quoter/companies/

Returns a list of display company names.

Query params

- toolkit - required (FEX, TERM, MEDSUPP, HI, IUL)

Success response (full fields)

```

[
  "AIG (Guaranteed Issue)",
  "Sentinel (New Vantage)"
]

```

Questionnaire Endpoints

These endpoints support condition/drug lookup and questionnaire traversal for underwriting item Q&A.

POST /questionnaire/search/drug/

POST /questionnaire/search/condition/

Searches by name and returns top matches.

Request body

```

{
  "name": "lisinopril"
}

```

Response

```

[
  "Lisinopril",
  "Lisinopril/HCTZ"
]

```

POST /questionnaire/traversal/

Initializes or resumes a traversal session.

Request body

```

{
  "toolkit": "FEX",
  "name": "Lisinopril",
  "is_drug": true
}

```

Optional:

- session_id to resume an existing traversal session

Success response (full fields)

All traversal endpoints (/traversal/, /traversal/answer/, /traversal/prev/, /traversal/edit/, /traversal/<session_id>/) return the same shape.

Top-level fields:

- has_next (Boolean)
- has_prev (Boolean)
- answer (List[String] | null) - current question answer
- name (String)
- is_drug (Boolean)
- question (TraversalQuestion)
- underwriting_items (List[TraversalUnderwritingItem])
- session_id (UUID String)

TraversalQuestion fields:

- input_type (String) - one of RADIO, DATE, DOUBLE_DATE, DROPDOWN, YES_NO
- type (String) - question type (examples: indication, second_level_indication, question_with_date, last_action, date_of_diagnosis, dosage, units_of_insulin, variable_dosage_dependent, age_dependent, yes_no)

- text (String)
- options (List[String], present for RADIO/DROPDOWN types)
- currently_text (String, present for last_action)
- currently_checkbox (Boolean, present for last_action)
- text2 (String, present for DOUBLE_DATE last-action prompts)
- type2 (String, present for DOUBLE_DATE last-action prompts)

TraversalUnderwritingItem fields:

- answer (String)
- type (String)
- question (String, optional)

```
{
  "has_next": true,
  "has_prev": false,
  "answer": null,
  "name": "Lisinopril",
  "is_drug": true,
  "question": {
    "input_type": "RADIO",
    "type": "indication",
    "text": "Indication",
    "options": ["High Blood Pressure (Controlled)"]
  },
  "underwriting_items": [],
  "session_id": "uuid"
}
```

POST /questionnaire/traversal/answer/

Saves answer(s) for the current question and advances if possible.

Request body

```
{
  "session_id": "uuid",
  "answer": ["current"]
}
```

Success response (full fields)

Same full traversal response schema documented under POST /questionnaire/traversal/.

POST /questionnaire/traversal/prev/

Moves the pointer back one step.

Request body

```
{
  "session_id": "uuid"
}
```

Success response (full fields)

Same full traversal response schema documented under POST /questionnaire/traversal/.

POST /questionnaire/traversal/edit/

Resets pointer to the beginning of the traversal session.

Request body

```
{
  "session_id": "uuid"
}
```

Success response (full fields)

Same full traversal response schema documented under POST /questionnaire/traversal/.

GET /questionnaire/traversal/<session_id>/

Fetches current traversal session state.

Success response (full fields)

Same full traversal response schema documented under POST /questionnaire/traversal/.

Building underwritingItems from Questionnaire Endpoints

Use questionnaire traversal responses to build each element in the `underwritingItems` array.

Step-by-step

1. Call POST /questionnaire/traversal/ with:

```
{
  "toolkit": "FEX",
  "name": "Lisinopril",
  "is_drug": true
}
```

2. Read the response:

- session_id identifies this questionnaire session
- question is the current question to display
- underwriting_items is the current cumulative answered-question list for this session
- has_next indicates whether more questions remain

3. For each question, call POST /questionnaire/traversal/answer/:

```
{
  "session_id": "<session_id>",
  "answer": [ "<selected_answer>" ]
}
```

Notes:

- answer is always an array
- For single-answer prompts, send one value (example: ["current"])
- For prompts that require two values, send both in order (example: ["current", "2021-07-05"])

4. Repeat until has_next is false.

5. Build one final quoter underwritingItems element from that completed session:

- name comes from traversal response name
- type is:
 - Drug when is_drug = true
 - Health Condition when is_drug = false
- answers comes from traversal response underwriting_items
- include hospitalization fields as needed by your client workflow

Final object shape per item

```
{
  "name": "Lisinopril",
  "type": "Drug",
  "hospitalization": false,
  "hospitalizationReason": null,
  "answers": [
    {
      "answer": "High Blood Pressure (Controlled)",
      "type": "indication"
    },
    {
      "answer": "current",
      "type": "time_dependent"
    },
    {
      "answer": "2021-07-05",
      "type": "first_fill"
    }
  ]
}
```

Build full underwritingItems array

- Repeat the same traversal flow for each selected drug/condition
- Append each completed item to underwritingItems
- Send the array to POST /quoter/ as underwritingItems

```
{
  "underwritingItems": [
    {
      "name": "Lisinopril",
      "type": "Drug",
      "hospitalization": false,
      "hospitalizationReason": null,
      "answers": [
        { "answer": "High Blood Pressure (Controlled)", "type": "indication" },
        { "answer": "current", "type": "time_dependent" },
        { "answer": "2021-07-05", "type": "first_fill" }
      ]
    },
    {
      "name": "High Blood Pressure (Controlled)",
      "type": "Health Condition",
      "hospitalization": false,
      "hospitalizationReason": null,
      "answers": [
        { "answer": "current", "type": "time_dependent" }
      ]
    }
  ]
}
```

Integration Notes

- Use exact field casing expected by each endpoint:
 - Quoter endpoints mostly use camelCase (`faceAmount`, `underwritingItems`, `paymentType`)
 - Simplified quoter uses snake_case fields (`face_amount`, `underwriting_items`, etc.)
- For underwriting quote accuracy, prefer full DOB (month, day, year) over age-only input.
- If `effectiveDate` is supplied, it cannot be in the past.
- Keep API keys server-side only; never embed in browser/mobile clients.

Support

For API key issuance, revocation, usage review, and environment setup, contact the ITK platform team.